Validation and AI Use

> **Saved July 31, 2026 assessment: watch/defer.**
> Low / base / high: $78.52 / $119.60 / $181.30; dated close: $308.91.
> $125.23 is the separate retrospective base, not this dated valuation.

PDF export note: These PDFs are current exports of saved review drafts. References below to final PDFs remaining unmet mean independently reviewed submission versions and their original checkpoint provenance remain unresolved; the export itself does not establish review or submission. Complete historical records and conflicting dated figures are preserved as labeled.

Validation and AI Use — follow-up

Scope, dates and environment

Primary July31,2026end-of-day17:00EDT investment assessment, using16:00close. Recorded execution date is in tool/test/cold logs; this is not a historical precommit. Retrospective model and prior AI/validation records preserved in archive/retrospective_v1/. Existing_work unchanged by editing actions. Python3.13.1 and requirements unchanged; fresh environment used outside OneDrive for this rerun. No new commit/push/publication.

Load-bearing evidence and conventions

Source_ledger.csv identifies every reported opening field, share measurement date, source availability, proxy transformation and assumption. Apple filing indexacceptedJuly31 06:01:02 establishes cover-share availability before close. June27actuals anchor model;July31balances are interpolation estimates. Primary July30DGS10 4.68% downloaded; beta/ERP/debtcost remain assumed. Market quotes are dated secondary unadjusted closes; adjusted alternatives rejected. MSFTrelease downloaded; SEC directcurl403 while web tool returned filing/index, recorded without pretending all files downloaded. Dates_and_methods.md details prorating, working-capital balance proxies, cash-equivalentSBC, dilution proxy and bridge.

Validation register

Check	Actual method/evidence	Limitation/status
Revenue/revolver provenance	test_ledger_fact_assumption_separation; actual p29 operations locator and assumption text	Reported zero revolver not asserted
Selected baseline	Nondefault GM/R&D/capex/shares/mincash; recompute every OATdelta and WACCcell; compare cases.base	Regression tests execute; no default-base mixing
Cash reserve	Min_cash+5000 reduces surplus/equity5000 in both models	No hidden module-default floor
Reported source anchor	June27assets383266/liabilities275746; source inputs checked	Dated reported balances, not July31measured

Check	Actual method/evidence	Limitation/status
Forecast and stub	Existing checks extended to explicit opening/year labels; FCFF two routes; all cases	Q4calendar/seasonality are estimates
Elapsed flows/timing	Stubsplit sums to full Q4FCFF;57/365firsttime; five full years beyond; independentPV recomputation	No elapsedFY26whole-year cash retained
EV/equity	Estimated bridge balances; EV+surplus−debt; selectedreserve; no futurecash addition	Detailed incremental claims remain uncompleted
DCF boundaries/directions	WACC/g refusal and monotonicity, signed flows retained, reverse reprices308.91	Reverse fixes base bridge; future buyback changes cannot rewrite elapsed cash
Peer definitions	July31close, JuneendingTTM, GAAPEBIT/NI, correctbookdebt/cash sums	One peer; strategic-equity/earnings-quality limits disclosed, no fake peer range
Product	AppTest operates defaultdatedcontrols/reset, lower margin and real failure/recovery	No browser/human recording asserted
Frozen output	Both datedJSONpaths equal new engine; retrospectivecontract equals corrected engine	Regenerate both after edits
Published training and prior finance tests	Existing ABGroundedknownanswer, FCFF definition, accounting/refusal tests retained	Ordinary assisted validation, no human AI-off claim
Preservation	Bounded subprocess attempts original checksum manifest and every protected file	Timeout producesSKIP with exactpath, notPASS. Prior successful hash record is historical only
Cold run	Fresh venv; selected runtime/source fixtures copied, originals/archive excluded; compare generated contract hashes; local HTTP health	Any skipped original-content read remains unverified. No complete-copy or human-access claim

Final packaging outcomes are in outputs/package_tests.log and the two package engine logs; the earlier runtime-only cold evidence is in outputs/cold_run_log.txt. Failed/stalled attempts are retained in outputs/refresh_tests_first.log/second.log and original archived logs. First test attempts stalled on OneDrive-backed originals and old bytecode; old caches moved aside, no original evidence overwritten. A15second preservation-read boundary now yields an explicit skip if unavailable. No financial result changed merely to force a pass.

Human locked record and boundary

Unchanged supplied locked record contains a student prediction/AI-closure confirmation claim but lacks original immutable pre-run timestamp/commit. It is not authenticated here. Current model changes, tests and demonstration are AI-assisted ordinary validation. Original EditionA receipt remains unverified. No prediction, receipt, timestamp, human review or historical authorship has been fabricated.

Material AI-use register

Exposed surface: Codex, GPT-6; finer model version not exposed. Toolsfunctions.exec/exec_command/write_stdin andweb.run. Session work: code fixes, source inspection, dated modeling/assumptions, peer screening/calculation, regression tests and document/video-outline preparation. Newprimaryrate modified preliminary datedbase130.32 to final119.60. Assistant accepted baseline/cash fixes after regression calculations; modified date method to reportedopening+estimatedstub/bridge; rejected stale annual peers and unreconciled Alphabet metrics. These are assistant dispositions, not the student's independent decisions. Student review/dispositions still pending.

Remaining limitations and conditions

See completion_status.md: multi-peer distribution, detailed claims/normalized earnings, human source review, EditionA/locked record, finalPDFs/media/access/submission and any skipped checksum verification remain UNMET. The dated model itself is implemented; estimates do not become reported facts. Base119.60, low78.52,high181.30; terminalshare68.91%. Watch/defer at308.91 subject to stated assumptions, not a currentOctoberclaim.

Submission packaging check — October 19, 2026 (local)

Research and feature work stopped at the user's request. Final saved code is the existing engine plus dated_refresh.py, with corrected retrospective baseline/reserve handling. The last correction changed only a risk-free source test from exact float equality to a 12-decimal tolerance and closed its input file; no valuation formula changed.

Executed once using the already installed local environment (no installation or new cold run):

```
/tmp/apple-project-refresh-venv/bin/python -B analysis.py > outputs/package_analysis_run.log 2>&1
/tmp/apple-project-refresh-venv/bin/python -B dated_refresh.py > outputs/package_dated_run.log
2>&1
/tmp/apple-project-refresh-venv/bin/python -B -m unittest discover -s tests -v >
outputs/package_tests.log 2>&1
```

All three commands exited 0. Tests: 32 run in 16.181 seconds; 31 passed, 1 skipped. Exact skip: `OneDrive read unavailable; preservation NOT reverified: READ existing_work/lab06/dcf.py`. A non-fatal ResourceWarning remains for an unclosed ledger input in test_refresh.py; it is recorded, not concealed.

The preceding failed run is retained in outputs/refresh_tests_latest.log: 32 tests, one failure and one skip. Failure: test_refresh.RefreshChecks.test_risk_free_observation_and_wacc, `AssertionError: 0.046799999999999994 != 0.0468`. The tolerance correction addresses binary floating-point representation; financial inputs and results were unchanged.

Final saved dated low/base/high: $78.52 / $119.60 / $181.30, July 31, 2026 end-of-day assessment against $308.91 close. Retrospective low/base/high: $72.74 / $125.23 / $172.33. Root visible_output.json and outputs/dated/ are primary; outputs/retrospective/ remains separate. No October current valuation is claimed.

Earlier cold_run_log.txt is a runtime-only check: selected runtime/source fixtures copied to a fresh environment; original evidence/archive excluded and preservation explicitly skipped. It ran the then-current 29 tests (28 passed, 1 skipped), regenerated matching contracts and received local HTTP health 200. It predates the last test additions/correction and is not a final full-repository cold run. No cold run was repeated for packaging.

Stalled prior Python test/read processes and git-status reads were terminated. Git status could not complete against OneDrive; no claim of a complete VCS diff inventory is made. Saved file pointers and scoped hashes are in docs/frozen_manifest.csv. An initial broader manifest read also stalled on OneDrive and was stopped; the final manifest intentionally hashes only selected active, readable files. It is not a complete repository freeze. No content reset/discard was performed. Original-content checksum verification remains incomplete; all peer/claim/human requirements in completion_status.md remain unmet. No human review, recordings, transcripts, publication or receipt was created.

Material source and definition table

Field / amount / status	Measurement / availability	Source, definition and verification
R26_AAPL_cash | cash | 39544.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_securities | securities | 106973.0 USD million | reported aggregation	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; 22855+84118; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_receivables | receivables | 31398.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_vendor_receivables | vendor_receivables | 27509.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_inventory | inventory | 11092.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_other_current_assets | other_current_assets | 17420.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_ppe | ppe | 51431.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances

Field / amount / status	Measurement / availability	Source, definition and verification
R26_AAPL_other_assets | other_assets | 97899.0 USD million | reported aggregation	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; 20342 intangible +77557 other noncurrent; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_payables | payables | 64525.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_deferred_revenue | deferred_revenue | 9538.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_other_liabilities | other_liabilities | 117339.0 USD million | reported aggregation	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; 62259 current +55080 noncurrent; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_commercial_paper | commercial_paper | 1997.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_debt | debt | 82347.0 USD million | reported aggregation	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; 11007 current +71340 noncurrent; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_revolver | revolver | 0.0 USD million | assumed	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	Analyst convention; PDF p2 balance sheet; 10-Q balance sheet; No modeled draw/capacity; not a reported zero; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances
R26_AAPL_equity | equity | 107520.0 USD million | reported	2026-06-27; available 2026-07-30 release; 10-Q 2026-07-31 06:01:02 accepted	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; PDF p2 balance sheet; 10-Q balance sheet; None; Primary statement read; June27 opening reconciles383266 assets/275746 liabilities/107520 equity; not July31 measured balances

Field / amount / status	Measurement / availability	Source, definition and verification
R26_AAPL_basic_cover_shares | basic_cover_shares | 14594.18 million shares | reported	2026-07-17; available 2026-07-31 before market open	https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm; Cover / statements of operations; Cover shares /1million; held flat as July31 estimate; Verified web filing; filing index accepted July31 06:01:02
R26_AAPL_quarter_basic | quarter_basic | 14656.11 million shares | reported	Q3 FY2026 weighted average; available 2026-07-31 before market open	https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm; Cover / statements of operations; Thousands to millions; Verified web filing; filing index accepted July31 06:01:02
R26_AAPL_quarter_diluted | quarter_diluted | 14714.676 million shares | reported	Q3 FY2026 weighted average; available 2026-07-31 before market open	https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm; Cover / statements of operations; Thousands to millions; Verified web filing; filing index accepted July31 06:01:02
R26_claim_proxy | diluted_common_equivalents | 14652.746 million common equivalents | estimate	2026-07-31 estimate; available Model estimate; not a reported count	https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm; Cover + Q3 EPS share note; 14594.180+(14714.676-14656.110); no future issuance; future SBC treated as cash compensation; Formula checked; not exact fully diluted July31 claims
R26_AAPL_ttm_ebit | TTM EBIT | 154859 USD million | calculated from reported	TTM to2026-06-27; available Latest quarterly release July30	https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm; Income statement plus FY25 10-K p29; 133050+122432-100623; Consistent GAAP operating-income definition; includes temporary tariff benefit
R26_AAPL_ttm_ni | TTM net income | 128930 USD million | calculated from reported	TTM to2026-06-27; available Latest quarterly release July30	https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm; Income statement plus FY25 10-K p29; 112010+101464-84544; GAAP net income, not adjusted EPS
R26_AAPL_close | unadjusted regular-session close | 308.91 USD/share | secondary dated market observation	2026-07-31 16:00 EDT; available 2026-07-31 close; retrieved during this session	https://finance.yahoo.com/quote/AAPL/history/?fr=sycsrp_catchall; July31 row Close (not Adj Close); Used same close date across target and peer; Web price row located; FinanceCharts corroboration; MSFT adjusted463.85 and AAPL adjusted308.64 rejected
R26_MSFT_close | unadjusted regular-session close | 464.72 USD/share | secondary dated market observation	2026-07-31 16:00 EDT; available 2026-07-31 close; retrieved during this session	https://uk.finance.yahoo.com/quote/MSFT/history/?ltr=1; July31 row Close (not Adj Close); Used same close date across target and peer; Web price row located; FinanceCharts corroboration; MSFT adjusted463.85 and AAPL adjusted308.64 rejected
R26_MSFT_shares | shares | 7427 million shares | reported	2026-06-30 point balance/shares or FY ended June30 earnings; available 2026-07-29 release	https://www.microsoft.com/en-us/investor/earnings/fy-2026-q4/press-release-webcast; Balance sheet / FY2026 income statement; sources/msft_fy26_release.html; June30 common outstanding; held flat as date proxy; Primary release inspected and downloaded; uses latest June-ending 12-month period

Field / amount / status	Measurement / availability	Source, definition and verification
R26_MSFT_ebit | ebit | 155237 USD million | reported	2026-06-30 point balance/shares or FY ended June30 earnings; available 2026-07-29 release	https://www.microsoft.com/en-us/investor/earnings/fy-2026-q4/press-release-webcast; Balance sheet / FY2026 income statement; sources/msft_fy26_release.html; GAAP; Primary release inspected and downloaded; uses latest June-ending 12-month period
R26_MSFT_net_income | net_income | 133749 USD million | reported	2026-06-30 point balance/shares or FY ended June30 earnings; available 2026-07-29 release	https://www.microsoft.com/en-us/investor/earnings/fy-2026-q4/press-release-webcast; Balance sheet / FY2026 income statement; sources/msft_fy26_release.html; GAAP; Primary release inspected and downloaded; uses latest June-ending 12-month period
R26_MSFT_debt | debt | 40294 USD million | reported aggregation	2026-06-30 point balance/shares or FY ended June30 earnings; available 2026-07-29 release	https://www.microsoft.com/en-us/investor/earnings/fy-2026-q4/press-release-webcast; Balance sheet / FY2026 income statement; sources/msft_fy26_release.html; 9227+31067; Primary release inspected and downloaded; uses latest June-ending 12-month period
R26_MSFT_cash_securities | cash_securities | 76843 USD million | reported aggregation	2026-06-30 point balance/shares or FY ended June30 earnings; available 2026-07-29 release	https://www.microsoft.com/en-us/investor/earnings/fy-2026-q4/press-release-webcast; Balance sheet / FY2026 income statement; sources/msft_fy26_release.html; 20935+55908; Primary release inspected and downloaded; uses latest June-ending 12-month period
R26_rf | risk-free proxy | 0.0468 annual decimal | reported observation	2026-07-30 observation; available Prior-business-day observation used for July31 end-of-day cutoff; current retrieval not archival receipt	https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10&cosd=2026-07-29&coed=2026-07-30; sources/dgs10_july2026.csv; Federal Reserve H.15 via FRED; 4.68 percent /100; Downloaded primary CSV; historical vintage receipt not created
R26_wacc | WACC | 0.10605484841351381 annual decimal | estimate	2026-07-31 estimate; available Model estimate	dated_refresh.py; WACC components; Rf4.68% + beta1.10*ERP5.5%=Ke10.73%; debt cost5%, shield21%; book gross debt/(close*latest basic count+debt); Rf observed; beta/ERP/debt cost are assumptions, not invented sourced estimates
R26_q4_growth | FY26 Q4 revenue growth | 0.1 YoY decimal | assumption	FY26 Q4 estimate; available As-of July31 forecast judgment	https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf; Prior-year Q4 = FY25 annual416161 - FY25 nine-month313695; FY26 Q4=102466*1.10; FY26 annual=364357+112712.6=477069.6; FY26 first nine months anchored to reported results;10% Q4 forecast is judgment, consistent with dated secondary guidance reports
R26_q4_margin | FY26 Q4 gross margin | 0.475 gross profit/revenue | assumption	FY26 Q4 estimate; available As-of July31 forecast judgment	https://www.apple.com/newsroom/2026/07/apple-reports-third-quarter-results/; Q3 release margin/refund disclosure; forward margin is judgment; Below reported Q3 50.1% and approximate ex-refund48.1%; not five-year management guidance; No invented primary guidance transcript; later annual margins48/48.5/49/49/49% are assumptions

Field / amount / status	Measurement / availability	Source, definition and verification
R26_date_bridge | July31 cash | 39879.35220857469 USD million | estimate	2026-07-31 estimate; available Calculated, not reported	dated_refresh.py; June27-to-Sept26 stub roll-forward; June27 cash +34/91 of modeled Q4 cash change; all balance lines interpolate on same basis; Estimated balance sheet reconciles; no exact July31 balance required or asserted

Material assumption challenges

Driver / value	Basis and challenge	Evidence that changes it / attribution
dated_date; July31,2026 EOD	LatestJuneperiodfilings beforecutoff; actualtimingroll; LinearQ4cashflows/withinquarterbalanceproxy	Quarterlyseasonaldata/capitalallocation showing materialdeviation; Current AI-assisted analytical challenge; independent student disposition pending
dated_q4; 10%growth/47.5%GM	Forecastjudgments after9mactuals; Memory/supply/refundcosts maychangequarter	Nextactualquarter versus forecast112712.6revenue; Current AI-assisted analytical challenge; independent student disposition pending
dated_annual; FY27–31 growth8/6/5/4/3%;GM48/48.5/49/49/49%	Analystfade,notmanagementguidance; Highpersistentefficiency maynotlast	Twoquartersgrowth<6%or exrefundGM<47%; Current AI-assisted analytical challenge; independent student disposition pending
dated_cash; 39879.35220857469	Juneactual +34/91stubcashchange; Unknownactualdistribution/workingcapitalti ming	Nextreportcash/cashflowreconciliation; Current AI-assisted analytical challenge; independent student disposition pending
dated_shares; 14652.746	July17basic+58.566mdilutionproxy; Notfullyclaim-specific;cashcompassumption	Actualawards/issuance/repurchase filings; Current AI-assisted analytical challenge; independent student disposition pending
dated_wacc; 0.10605484841351381	Rf4.68%primary;beta1.10ERP5.5%debtco st5%assumed; Risk/spread/ERP assumptions	Datedbeta/spread/premium evidence; Current AI-assisted analytical challenge; independent student disposition pending
dated_terminal_ROIC; 0.6428890384973598	g divided by terminalreinvestment/NOPAT; Demanding sustainedincrementalreturn	Highercapitalintensity/lower sustainableoperatingreturns; Current AI-assisted analytical challenge; independent student disposition pending
dated_peer; MSFTonly	GAAPmatchedJuneTTM/commonJulyclos e; Onepeer notdistribution; strategicinvestments/GAAPgainlimits	Secondqualifyingpeer withclaims/earningsnormalization; Current AI-assisted analytical challenge; independent student disposition pending

Methods and date conventions

Dated valuation and preserved retrospective model

Primary decision date: **July31,2026 end of day (17:00EDT)**; market reference **July31 regular-session close16:00EDT**, AAPL308.91. This is a historical dated investment assessment, not a newly asserted October19 price target. Cutoff excludes later earnings and quotes. Apple releaseJuly30 and 10-Q acceptedJuly31 06:01:02 were public; filing index is in the source ledger. Microsoft June30 results releasedJuly29; Alphabet June30 results/filing also preceded cutoff. July30 DGS10 prior-business-day observation4.68% is downloaded from primary FRED/H.15. No fabricated historical receipt or vintage snapshot.

The prior September27,2025 illustration is preserved as code and original outputs in archive/retrospective_v1/ and regenerated under outputs/retrospective/. It still gives72.74/125.23/172.33. No time label was merely renamed.

June27 reported opening: assets383266, liabilities275746, equity107520; cash39544; securities106973; gross CP+term debt84344. The opening cash-flow/reinvestment projection uses these actual balances, including97899 intangible/other noncurrent assets and117339 other liabilities; it does not retain the obsolete FY25 opening. Zero revolver is a model convention, not a reported fact.

FY26 Q4 sales forecast = (416161−313695)*1.10 =112712.6. Add actual FY26nine-month sales364357 to obtain FY26 estimated annual sales477069.6. Q4 gross margin47.5% and10% growth are analyst judgments, with near-term supply/memory/refund risks. Annual forecast FY2027–2031 grows8/6/5/4/3%, margin48/48.5/49/49/49%. R&D9.5%, capex3%, operating tax18% retained. Existing statement engine is extended only to accept an opening balance, fiscal-year labels and a fractional first period. It computes a91-day Q4 from June27 toSeptember26; depreciation, interest and fixed distributions are prorated, revenue-based expenses/capex use actual forecast-quarter sales. Balance proxies use estimated annual sales and Q4 cost mix, not one quarter of sales multiplied by annual balance ratios. These proxies and flat other liabilities/securities are explicit estimates.

July31 is34/91 through that quarter. All balance lines interpolate between June27 reported and projected FY26end, maintaining balance-sheet articulation. Estimated July31 cash is39,879.35; gross debt remains84344 by a refinancing/held-flat assumption, not a reported July31 balance. This is a defensible between-report estimate, not a requirement to know exact current amounts. A separate latest-reported-only bridge check is exported.

DCF excludes9,006.10 elapsed estimated FCFF and includes only15,098.47 remaining FY26 FCFF. Remaining stub arrivesSeptember26 at57/365 years. Five full years FY2027–2031 follow, ending on assumed future fiscal Saturdays; discount times are[0.15616438356164383, 1.1534246575342466, 2.16986301369863, 3.1671232876712327, 4.164383561643835, 5.161643835616438]. Terminal cash flow is FY2031 FCFF*(1+g); terminal value is at FY2031end, not five years from an obsolete anchor. Linear within-quarter cash timing is an approximation. No completed FY26 cash flow is counted again.

EV1,719,915.09 + estimated July31 surplus116,852.35 − debt84344 = equity1,752,423.44. Surplus = securities106973 + max(estimated cash−selected operating reserve30000,0). Only valuation-date estimated surplus is added; future cash is never added. Higher selected min_cash lowers surplus. Other claims0 is an explicit incremental-claim assumption; other liabilities remain operating aggregates with lease rent/tax effects in forecasts. Detailed non-operating tax/legal/award claim valuation remains uncompleted and is a material risk.

Shares: reported July17 common14594.180million, plus estimated existing dilution58.566million (Q3 diluted14714.676 minus basic14656.110 weighted-average shares) ->14,652.746million estimated common equivalents. This is a proxy, not an exact reported fully diluted date count. Future SBC remains an operating expense/cash equivalent, with no SBC addback/new issuance; buybacks are distributions without assumed per-share accretion. Basic result120.08 is separately visible. A claim-specific award valuation is not completed.

WACC10.6055%: Rf4.68%, assumed beta1.10, assumed ERP5.5% ->Ke10.73%; assumed marginal debt cost5%, shield21%. Gross book debt/(July31close*July17basic shares+debt) weight1.84%. Do not call beta, ERP or debt cost verified market observations. Terminalg2.5%, implied continuing ROIC64.29%; terminal share68.91%. Low/high WACCs11%/9% are causal risk judgments. Signed negative cash flows are retained; invalid perpetuities refuse.

FCFF mapping and historical normalization remain as before: EBIT*(1−operating tax)+PP&E depreciation+residual operating D&A−PP&E capex−other reinvestment−Δoperating NWC. Other D&A matched by reinvestment; cash routeCFO+CFI+NOPAT−NI. No debt repayment, buybacks or dividends in FCFF. FY24 reported tax29749 excludes identified one-time StateAid10246 for normalized19503; FY25 tax20719 excludes benefit486 for normalized21205 and NI111524. Reported balance sheets are not overwritten by normalization.

Locked record: limitations before reading

The following supplied record is reproduced in full without correcting its wording or numbers. Its claims that the student confirmed AI closure and real exchanges are historical assertions in that record, not newly verified facts. Original immutable prediction timestamp/commit, authenticated AI-off execution and numerical partner review are not established. The September 29 execution timestamp is not the original prediction timestamp.

The record concerns FY2026-2030 growth and supplementary FCFE. Its $18,821.48 profit change, $13,592.81 FCFE change and $9.65 per-share change are historical results, not the dated July 31 FCFF sensitivity. The dated +2 percentage-point growth result is +$19,761.69 million profit, +$14,305.46 million FCFF and +$9.13 per share. No conflicting figures are silently replaced.

Locked Changed-Input Record — Apple

Status: The student confirmed independent saving with AI closed and actual before-run review in both roles before scenarios ran. The original save timestamp/timezone or Git commit was not supplied; no earlier time is invented.

Prediction supplied by the student

> I predict that increasing Apple's revenue growth assumptions from 15%, 8%, 6%, 5%, and 4% to 17%, 10%, 8%, 7%, and 6% for FY2026–FY2030 will increase FY2030 operating profit by approximately 10%, or $19,500 million. I expect FCFE to increase by roughly 8%, or $11,700 million, and value per share to increase by approximately $10. Higher growth compounds over the five years, increasing revenue and profit. However, additional investment and working capital could offset some of the cash flow increase. All other independent assumptions will remain at their base values.

Input: annual revenue growth, +2 percentage points in each FY2026–FY2030 year. Gross margin and all other independent assumptions remain at base for this predicted run. The approximate dollar and percentage predictions above are retained as supplied, without adjusting either.

Before-run confirmation

The student replied "yes, yes real exchanges" when asked to confirm independent saving with AI closed and actual partner review of predictions, ranges, units and one-input-at-a-time changes in both roles. The discussions are preserved in partner_discussion_notes.md. The exact original save timestamp and specific numerical partner checks were not supplied.

Actual result and reconciliation

Output change	Prediction	Actual	Actual − prediction
Operating profit ($m)	+19,500.00	+18,821.48	-678.52
FCFE ($m)	+11,700.00	+13,592.81	+1,892.81
Value ($/share)	+10.00	+9.65	-0.35

Operating profit rose 9.66%, slightly below the predicted 10%; FCFE rose 9.28%, above the predicted 8%. The input change compounded revenue upward; fixed expense ratios preserved the operating-profit relationship. Additional tax, working capital and capex offset part of the gain, while depreciation added back cash. The exact offsets produced a larger FCFE gain than the rough prediction. The valuation increase of $9.65/share was close to the predicted $10 and reflects the whole cash-flow path and terminal value. Full statement trace: report (Submission/Lab_11_Apple.md).

Analysis execution timestamp (not the original prediction timestamp): 2026-09-29T21:42:12.039652+00:00

Actual final validation results

Actual final validation results

Submission packaging check — October 19, 2026 (local)

Research and feature work stopped at the user's request. Final saved code is the existing engine plus dated_refresh.py, with corrected retrospective baseline/reserve handling. The last correction changed only a risk-free source test from exact float equality to a 12-decimal tolerance and closed its input file; no valuation formula changed.

Executed once using the already installed local environment (no installation or new cold run):

```
/tmp/apple-project-refresh-venv/bin/python -B analysis.py > outputs/package_analysis_run.log 2>&1
/tmp/apple-project-refresh-venv/bin/python -B dated_refresh.py > outputs/package_dated_run.log
2>&1
/tmp/apple-project-refresh-venv/bin/python -B -m unittest discover -s tests -v >
outputs/package_tests.log 2>&1
```

All three commands exited 0. Tests: 32 run in 16.181 seconds; 31 passed, 1 skipped. Exact skip: `OneDrive read unavailable; preservation NOT reverified: READ existing_work/lab06/dcf.py`. A non-fatal ResourceWarning remains for an unclosed ledger input in test_refresh.py; it is recorded, not concealed.

The preceding failed run is retained in outputs/refresh_tests_latest.log: 32 tests, one failure and one skip. Failure: test_refresh.RefreshChecks.test_risk_free_observation_and_wacc, `AssertionError: 0.046799999999999994 != 0.0468`. The tolerance correction addresses binary floating-point representation; financial inputs and results were unchanged.

Final saved dated low/base/high: $78.52 / $119.60 / $181.30, July 31, 2026 end-of-day assessment against $308.91 close. Retrospective low/base/high: $72.74 / $125.23 / $172.33. Root visible_output.json and outputs/dated/ are primary; outputs/retrospective/ remains separate. No October current valuation is claimed.

Earlier cold_run_log.txt is a runtime-only check: selected runtime/source fixtures copied to a fresh environment; original evidence/archive excluded and preservation explicitly skipped. It ran the then-current 29 tests (28 passed, 1 skipped), regenerated matching contracts and received local HTTP health 200. It predates the last test additions/correction and is not a final full-repository cold run. No cold run was repeated for packaging.

Stalled prior Python test/read processes and git-status reads were terminated. Git status could not complete against OneDrive; no claim of a complete VCS diff inventory is made. Saved file pointers and scoped hashes are in docs/frozen_manifest.csv. An initial broader manifest read stalled and was stopped; the final manifest is a selected-file snapshot, not a complete repository freeze. No content reset/discard was performed. Original-content checksum verification remains incomplete; all peer/claim/human requirements in completion_status.md remain unmet. No human review, recordings, transcripts, publication or receipt was created.

All unresolved requirements

Completion status — finance refresh implemented; final submission NOT complete

Completed finance/code changes

Source ledger revenue locators/revolver classification fixed. Selected-baseline and selected-reserve bugs fixed with tests. START-HERE/README point to implemented work. Original retrospective model preserved separately; default app now presents dated July31assessment. Source inputs are latest June-period reports available by July31; estimated between-report balances are expressly labeled, not unknowable exact amounts demanded. Remaining FY26stub and five full yearsFY2027–2031 feed required FCFF/WACC, bridge, sensitivities, rankings, table and dated reverseDCF. Dated low/base/high78.52/119.60/181.30; cutoffJuly31end-of-day; close308.91. One matching-window enterprise/equity peer comparison is calculated from sourced inputs. See outputs/dated/ and dates_and_methods.md.

Requirements still UNMET — disclosure is not completion

Requirement/deliverable	Exact remaining work
Broader comparable-company triangulation	Only MSFT admitted. No multi-peer valuation distribution/leave-one-out robustness. Complete AlphabetA/B/C common-equity, convertible preferred/capped-call and non-operating investment reconciliation, common-earnings normalization; or source another qualifying peer under policy. Current single-peer EV/EBIT and equity/NI calculations themselves ARE implemented.
Fully disaggregated equity claims and earnings quality	Current zero incremental claims/operating other liabilities and58.566m award-dilution proxy are assumptions. Detailed non-operating tax/legal/award valuation and strategic investment/GAAP normalization are NOT completed. They do not require impossible exact-date balances. Latest filings plus disclosed estimates should be used.
Independent student finance review and AI disposition	Student must inspect sources and calculations, challenge terminal efficiency/WACC/cash/claims, and record real accept/modify/reject reasons. No human review claimed.
Original EditionA checkpoint	Original PDF, first successful Brightspace receipt/time and any required matching freeze/approved recovery remain unverified. Preserved Markdown is not proof.
Locked human AI-off changed-input record	Original immutable prediction timestamp/commit and authenticated AI-off execution absent. Current tests are ordinary AI-assisted validation; instructor-approved independent completion/recovery needed.

Requirement/deliverable	Exact remaining work
Full fresh preservation-content check	Existing_work was not edited. If OneDrive blocks reading a protected original, bounded check SKIPS and fresh checksum verification remains UNCOMPLETED. Final packaging check skipped existing_work/lab06/dcf.py after 15 seconds; see outputs/package_tests.log. Prior successful record is not substituted for a new result.
Final PDFs and real media	Reviewed memo≤2pages, Research-Evolution and Validation-and-AI-Use PDFs; your personal Videos1≤3min,2≤5min,3≤5min; corrected transcripts from actual recordings. No synthetic media or fake transcripts created.
Final submission/access	Stable repository/videoURLs, real durations/timestamps, logged-out access check, frozen package and final Brightspace receipt. No push/publication performed.

The dated-refresh requirement is implemented as a reported-data-plus-estimate investment valuation, not merely renamed old timing. An October19current recommendation and exact-date audited balance sheet are neither produced nor required to make this July31assessment. No claim of complete submission readiness. Sources, estimates and unfinished claim/peer-quality work are distinct.

Evidence pointers

analysis.py and outputs/retrospective/: corrected selected-baseline illustration; dated_refresh.py and outputs/dated/: current dated assessment. outputs/visible_output.json is the dated primary contract. Source ledger, peer addendum and bridge_reported_vs_estimated.csv explain provenance. tests/test_refresh.py covers fixes and dated timing/bridge. Test/cold logs record actual outcomes, including skips. Historical first-run logs/documents are preserved under archive/retrospective_v1/. Existing_work untouched; no new human history asserted.

Final packaging result

Both engines executed successfully; 32 tests: 31 passed, one preservation skip. The preceding float-equality failure and non-fatal ledger ResourceWarning are disclosed in outputs/validation_results.md. No new cold run was attempted. Saved code/results are executable; the unmet requirements above are not represented as completed.